

AMIT RAJ REDDY DHARAM

903-990-6812 | reddy.amitraj@gmail.com | linkedin.com/in/arrd19/ | ames-zero.github.io | Tempe, AZ

SUMMARY

Software engineer focused on building AI-powered backend systems, cloud deployment, end-to-end solutions, and agentic AI workflows using AI coding assistants. Seeking software developer (AI/ML), AI engineer, data science, and data engineer positions starting May 2026.

EDUCATION

Masters in Computer Science, Arizona State University (Tempe, AZ, United States of America) **May 2026**

Coursework: Artificial Intelligence, Statistical Machine Learning, Data Mining, Cloud Computing, Distributed Systems, IOT

Bachelors in Computer Science and MSc in Mathematics, BITS Pilani (Hyderabad, TG, India) **May 2023**

Coursework: Data Structure and Algorithms (C/C++), OOP (Java), Deep Learning (Python), Databases(SQL)

TECHNICAL SKILLS

Languages and Technologies: Python, Java, JavaScript, HTML, Go, C/C++, Flask, FastAPI, PostgreSQL, MongoDB, Neo4j

ML frameworks and libraries: PyTorch, TensorFlow, Pandas, PySpark, Scikit-Learn, NumPy, Matplotlib, OpenCV, Open3D, XGboost

Natural Language Processing (NLP): Hugging Face Transformers, LangChain, CrewAI, Google ADK, FAISS, Pinecone, BERT, OpenAI APIs

Developer tools: Amazon Web Services, Google Cloud Platform (GCP), Docker, Kubernetes, Grafana, GitHub, Terraform, GitHub Copilot

Concepts: Deep learning, AI cloud deployment, backend development and API design, generative AI, computer vision, data science

PROFESSIONAL EXPERIENCE

AI Data Science Analyst (Part-time) **Sept 2024 – June 2025**

Enterprise technology at Arizona State University

Tempe, AZ

- Implemented backend for a natural language conversational AI agent using **Python** and **FastAPI**, allowing users to query, filter, and display data from **NoSQL** and relational databases using conversational inputs, with query execution accuracy over 85%
- Integrated **LangChain** to orchestrate workflows and used **GraphRAG** with **Neo4j** for semantic query search. Deployed the system on **AWS EKS** with CI/CD pipelines, enabling scalable inference and achieving 30% faster response times for over 100 queries per session
- Designed a query optimization layer with caching using **Redis** and collaborated with other teams through code reviews and regular sprints to reduce database response times by 40% and improve system reliability.

Software Development Engineer - Machine Learning **July 2023 – June 2024**

TELUS Digital

India

- Finetuned open-source LLMs using **PyTorch** and **Hugging Face** on domain specific scientific datasets using PEFT-based methods for creating synthetic datasets. Increased manual question generation throughput by 50%, reducing generation time by 40%
- Instrumented ML services with latency and error measurements using **Grafana** dashboards alongside **AWS CloudWatch**, resulting in 30-40% performance gains through rigorous data-driven iteration, Redis cache implementation, and query optimization tactics
- Mentored interns through targeted guidance to test the LLM against key NLP benchmarks (MATH, MMLU), achieving +6% score improvements and 50% faster evaluations through automated pipeline workflows and efficient parallel processing

Software Development Intern - Machine Learning **Aug 2022 – July 2023**

TELUS Digital

India

- Prototyped advanced DeepSORT and ByteTrack algorithms for precise object bounding box tracking across video sequences, achieving 80% mAP object recognition and 75% IDF1 multi-object tracking accuracy, rigorously validated against MOT17 benchmarks
- Improved image segmentation precision by 23% across 15,000+ annotated frames by fine-tuning U-Net and Mask R-CNN architectures using the CityScapes dataset, reducing manual annotation time from 8 hours to 5 hours per dataset batch
- Designed **pytest** integration test strategies for **FastAPI/TorchServe** ML APIs serving 15+ projects, maintaining sub-200ms response times, 90%+ code coverage via **GitHub Actions** CI/CD automation, and JWT authentication security

PROJECTS

WardrobeAI (Recommendations and Image Generation) | *Winner Sunhacks, Sept 2025* **Project-Link**

- Built a robust **Python FastAPI** backend composed of 5+ microservices, integrating **MongoDB**, **AWS S3**, and CRM-style user profiles with **Google Gemini APIs** to provide virtual clothing try-ons and deliver personalized outfit suggestions across 3,000+ clothing items
- Coordinated a 4-person team to architect and deploy a sophisticated LinUCB recommendation engine integrated with a conversational chatbot, delivering highly personalized outfit suggestions tailored for 10+ occasions including birthdays, weddings, and formal events

Tom-Riddle's Diary (AI Chatbot Agent) | *Winner InnovationHacks, Apr 2025* **Project-Link**

- Engineered and integrated a Python FastAPI backend using **ChromaDB** (vector database) to enable long-term semantic memory and efficient information retrieval, reducing response time by 35% and supporting contextual AI reflections
- Integrated Google Gemini LLMs into a conversational backend pipeline, enabling natural language queries and persona-based tone generation, handling over 1,000 personalized interactions with >90% response relevance

ACTIVITIES AND LEADERSHIP

- Actively participated in hackathon teams to build end-to-end full-stack AI/ML applications, leveraging **GitHub Copilot**, debugging expertise, and problem-solving skills to deliver production-ready prototypes in 24-48 hours, securing multiple first-place wins
- Volunteered at OpportunityHack (ohack.org), proactively taking initiative to direct development teams with meticulous attention to detail while constructing robust websites and digital solutions for NPOs including the Heritage Square Foundation and NMTSA